

Co-Instruct++: Generalizable Image Quality Assessment Via Open-Ended Visual Comparison

Hanwei Zhu, Yuhan Luo, Baoliang Chen, Xi Zhang, Shiqi Wang, *Senior Member, IEEE*,
Yuming Fang, *Fellow, IEEE*, and Weisi Lin, *Fellow, IEEE*

Abstract—Image quality assessment (IQA) is commonly formulated as single-stimulus score prediction, even though human judgments are inherently comparative, context dependent, and often require explanation. Although comparative judgments can reduce the ambiguity of absolute scoring, existing comparative IQA methods are usually confined to fixed protocols and are not jointly designed for open-ended reasoning and scalar quality estimation. In this paper, we present Co-Instruct++, a unified large multimodal model (LMM) framework for generalizable IQA through open-ended visual comparison. We formulate IQA as multimodal comparative reasoning: a model compares multiple images, describes perceptual differences in natural language, and transfers the acquired comparison ability to quality rating. To realize this formulation, we construct a two-stage instruction-tuning dataset. Stage I provides comparative-quality-understanding instructions from Merge2Compare and Teach2Compare, covering image pairs and groups of three to six images under a coarse-to-fine grouping strategy. Stage II introduces score-guided comparative rating instructions that align relative judgments with rating supervision, thereby connecting open-ended comparison with perceptual quality prediction. The Co-Instruct++ series comprises Co-Instruct-Assistant for open-ended multi-image quality reasoning and Co-Instruct-Rater for rating-aware prediction. We further introduce MICBench_v2, a human-validated benchmark for multi-image quality reasoning that spans diverse question types, comparison granularities, distortion characteristics, and reasoning complexity. Experiments on established single-image, pairwise, multi-image, and scalar IQA benchmarks show that Co-Instruct++ outperforms open-source LMMs and remains competitive with strong proprietary systems. The dataset and models will be released at <https://github.com/h4nwei/Co-Instruct-Plus>.

Index Terms—Large multimodal model, visual quality comparison, image quality assessment.

1 INTRODUCTION

Image quality assessment (IQA) is a fundamental problem in visual computing, supporting content selection, restoration, compression, image generation, and other quality-sensitive applications. Classical IQA methods usually formulate the task as single-stimulus score prediction, in which each image is independently mapped to a scalar perceptual quality score [1]. More recently, large multimodal models (LMMs) have expanded IQA from numerical prediction to open-ended visual quality understanding, enabling models to answer natural-language queries and provide interpretable quality-related descriptions [2], [3], [4], [5]. This capability is important because practical quality evaluation often requires not only a score, but also comparison, selection, attribute-specific diagnosis, and explanation.

Despite this progress, most classical and LMM-based IQA methods remain centered on *single-stimulus evaluation*. This formulation implicitly assumes that perceptual quality can be assessed in isolation [6], whereas human quality judgments are often relative and context dependent. In subjective studies, observers may disagree about the absolute quality of an individual image but exhibit greater consistency

when comparing multiple images and identifying relative differences [7], [8]. As illustrated in Fig. 1, an isolated image can admit an ambiguous absolute evaluation, while comparison exposes relative artifacts, detail preservation, and perceptual trade-offs. This observation motivates a broader question: can IQA be modeled as open-ended comparative reasoning rather than solely as independent score prediction?

Existing comparative IQA methods only partially address this question. Many are restricted to predefined protocols, such as pairwise preference or forced-choice comparison, and therefore provide limited support for flexible user queries, multi-image reasoning, and natural-language explanation. Meanwhile, existing LMM-based IQA approaches mainly focus on single-image quality understanding or prompt-guided scoring, leaving the connection between comparative reasoning and scalar quality prediction underexplored. Consequently, comparison, reasoning, and rating are often treated as separate tasks, limiting scalability and weakening generalization across practical quality-evaluation scenarios.

To address this gap, we formulate IQA as an *open-ended multimodal comparative reasoning problem* and develop a unified framework that connects multi-image comparison, natural-language explanation, and perceptual quality estimation. To supervise this formulation, we construct Co-Instruct-359K, a two-stage instruction-tuning dataset with 359K question-answer pairs. Stage I provides comparative-quality-understanding instructions through Merge2Compare and Teach2Compare, covering image pairs and groups of three to six images under a coarse-to-fine grouping strategy. Stage II introduces *score-guided comparative rating* instructions that align relative quality judgments with rating supervision,

- Hanwei Zhu and Yuming Fang are with the School of Computing and Artificial Intelligence, Jiangxi University of Finance and Economics, Nanchang, Jiangxi, China. (e-mail: hanwei.zhu@my.cityu.edu.hk, fa0001ng@e.ntu.edu.sg).
- Yuhan Luo, Baoliang Chen, Xi Zhang, and Weisi Lin are with the College of Computing and Data Science, Nanyang Technological University, Singapore. (email: {baoliang.chen, xi.zhang, wslin}@ntu.edu.sg, yluo016@e.ntu.edu.sg).
- Shiqi Wang is with the Department of Computer Science, City University of Hong Kong, Kowloon, Hong Kong. (emails: shiqi.wang@cityu.edu.hk).

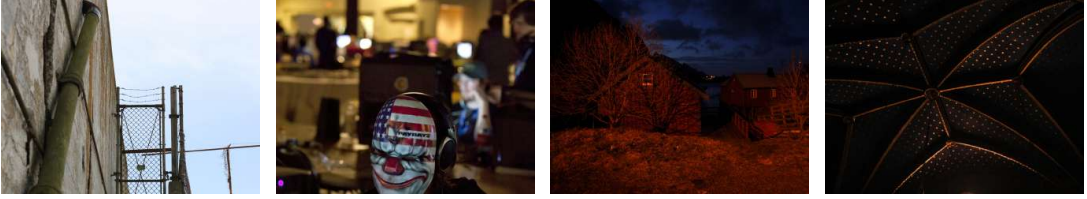
Images				
	(1) (2) (3) (4)			
Quality Evaluation (single image)	How is the exposure level of the image? Please choose among "high", "medium" and "low". 9 observers answered high 8 observers answered medium 13 observers answered low 6 observers answered medium 7 observers answered low 2 observers answered medium			
Quality Comparison (multiple images)	Please rank the exposure level among the four images. all 15 observers answered (1)>(2)>(3)>(4)		Which image has the highest exposure? all 15 observers answered (1)	

Fig. 1. **Motivation for open-ended visual quality comparison.** Absolute single-image quality assessment can be ambiguous, whereas comparative evaluation provides contextual cues for identifying relative perceptual differences.

thereby transferring open-ended comparison ability to perceptual quality prediction.

Built on this dataset, we develop the Co-Instruct++ series. Co-Instruct-Assistant learns open-ended multi-image quality reasoning from Stage I and supports ranking, selection, attribute-level comparison, and explanatory responses. Co-Instruct-Rater is initialized from the assistant and further optimized on Stage II with ordinal quality supervision and dataset-wise ranking consistency, enabling rating-aware prediction while preserving comparative reasoning ability. To support systematic evaluation, we further construct MICBench_v2, a human-validated benchmark for multi-image quality reasoning that spans diverse question types, comparison granularities, distortion characteristics, and reasoning complexity. Experiments on established single-image, pairwise, multi-image, and scalar IQA benchmarks show that Co-Instruct++ outperforms open-source LMMs and remains competitive with strong proprietary systems, supporting comparative reasoning as a principled basis for unified visual quality understanding.

Therefore, the main contributions of this work are four-fold:

- We introduce a unified formulation of IQA as open-ended multimodal comparative reasoning, connecting multi-image comparison, natural-language explanation, and perceptual quality estimation beyond fixed pairwise protocols.
- We construct Co-Instruct-359K, a two-stage instruction-tuning dataset with 359K question-answer pairs, where Stage I supports coarse-to-fine comparative understanding and Stage II bridges relative judgment with score-guided rating supervision.
- We develop the Co-Instruct++ series: Co-Instruct-Assistant for open-ended multi-image quality reasoning and Co-Instruct-Rater for rating-aware prediction with ordinal supervision and ranking consistency.
- We establish MICBench_v2 as a human-validated benchmark for multi-image quality reasoning and demonstrate the effectiveness of Co-Instruct++ across established single-image, pairwise, multi-image, and scalar IQA evaluation settings.

This work substantially extends our ECCV oral paper Co-Instruct [9], which introduced open-ended visual quality comparison as a problem setting, along three substantive dimensions. First, Co-Instruct-359K extends the data formulation to comparisons involving up to six images, explicitly covers both coarse- and fine-grained quality reasoning, and introduces score-guided rating supervision. Second, the proposed Co-Instruct++ series advances beyond comparison-only instruction tuning: Co-Instruct-Assistant supports open-ended comparative reasoning, and Co-Instruct-Rater transfers this capability to perceptual quality estimation. Third, MICBench_v2 provides a richer, human-validated evaluation of multi-image quality reasoning under greater comparison complexity. Together, these additions yield a unified framework that connects relative comparison, interpretable reasoning, and quality rating within a generalizable IQA system.

2 RELATED WORKS

In this section, we review prior studies from two perspectives: conventional IQA and quality ranking, and LMM-based approaches to visual quality understanding.

2.1 Conventional IQA and Quality Ranking

Prior to LMM-based quality understanding, IQA was studied primarily under two paradigms: score regression and relative ranking. Classical NR-IQA methods [10], [11], [12], [13] and subsequent deep models [14], [15], [16], [17] focused on learning quality-aware representations and mapping them to scalar quality scores, with robustness improved through domain adaptation, meta-learning, multi-task learning, and contrastive learning [16], [18], [19], [20], [21]. In parallel, ranking-based approaches [21], [22], [23], [24], [25] used pairwise preference supervision to alleviate the instability of absolute regression and better reflect the comparative nature of human judgment. Despite their effectiveness, these methods are limited to scalar predictions or pairwise decisions and therefore cannot support open-ended, multi-image, explanation-oriented quality understanding. Moreover, although paired comparison is widely recognized

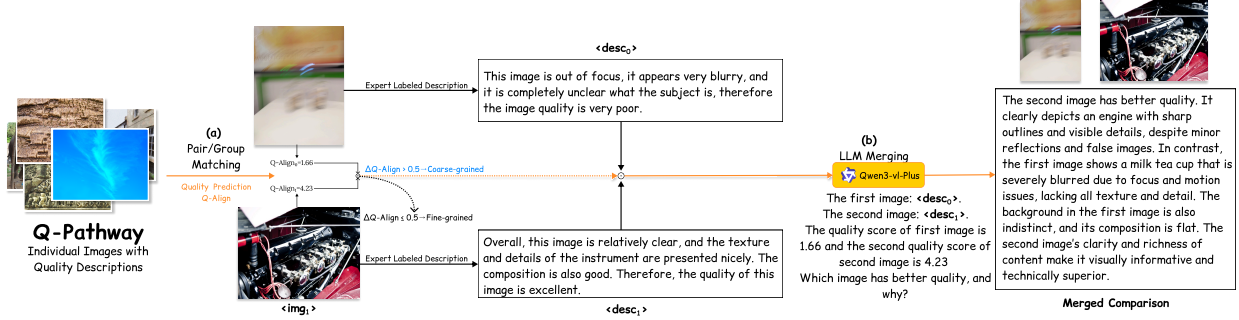


Fig. 2. Pipeline of Merge2Compare. (a) Candidate images are matched into groups according to predicted quality. (b) The associated human-authored quality descriptions and predicted scores are merged by an LMM to produce comparative instructions.

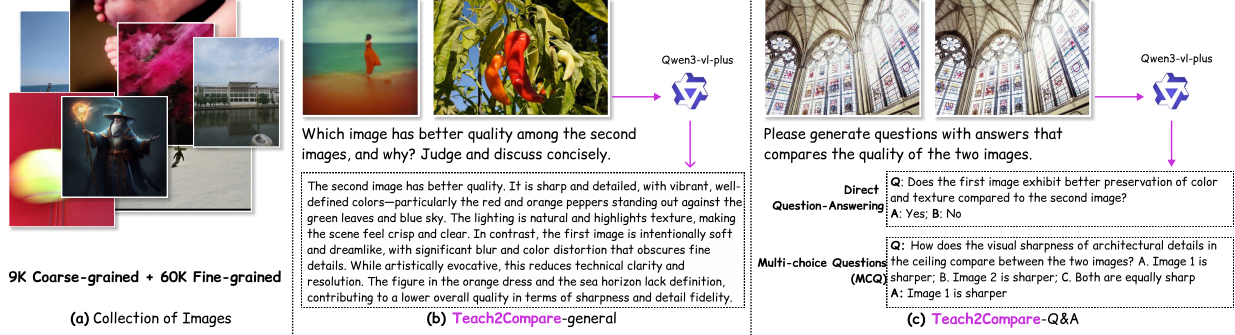


Fig. 3. Pipeline of Teach2Compare. A diverse image pool is first organized into comparison groups (a); a teacher LLM then produces free-form comparative reasoning (b) and structured question–answer supervision (c) for each group.

as a reliable protocol in subjective quality assessment [7], [8], its extension to richer multi-image reasoning settings remains underexplored. These limitations motivate IQA frameworks that integrate comparative reasoning, natural-language interpretation, and perceptual quality estimation.

2.2 LMM-Based IQA and Quality Reasoning

Recent work has introduced LMMs into IQA by extending quality assessment from scalar prediction to instruction-following visual quality understanding. Early studies, including Q-Bench, Q-Instruct, Q-Align, Compare2Score, Depict-QA, and DeQA-Score [2], [3], [4], [5], [26], [27], adapted vision–language models for quality scoring, description, and prompt-guided assessment. More recent efforts elicit explicit reasoning through supervised fine-tuning, reinforcement learning, or tool-assisted analysis, yielding more interpretable quality judgments [9], [28], [29], [30], [31]. Nevertheless, existing LMM-based IQA methods remain largely centered on single-image scoring or textual reasoning. They therefore do not fully address open-ended multi-image comparison or the transfer of comparative reasoning to scalar quality prediction; moreover, reasoning conducted solely in the textual space may discard continuous visual cues that are critical for quality perception [32], [33], [34]. In contrast, our work unifies open-ended comparative reasoning and quality estimation through multi-image comparative supervision, score-guided comparative rating learning, and a dedicated benchmark for multi-image quality reasoning.

3 CONSTRUCTION OF THE INSTRUCTION DATASET

We construct Co-Instruct-359K as a two-stage instruction corpus that supports both open-ended visual comparison and score-guided quality estimation. Stage I provides comparative-quality-understanding instructions for image pairs and groups of three to six images. It consists of two complementary pipelines, Merge2Compare and Teach2Compare, organized under a coarse-to-fine grouping strategy. Coarse-grained groups contain clear quality gaps and provide stable ranking cues, whereas fine-grained groups contain perceptually similar or distortion-sensitive samples that require attribute-aware discrimination. Stage II converts subjective quality scores into ordinal absolute-quality labels and uncertainty-aware relative-quality labels.

3.1 Stage I: Instructions for Comparative Quality Understanding

Stage I is designed to train the model to compare multiple images and explain perceptual differences in natural language. As summarized in Tab. 1, it contains four subsets: Merge2Compare-coarse, Merge2Compare-fine, Teach2Compare-coarse, and Teach2Compare-fine. These subsets cover both free-form comparative reasoning and question-driven comparison for image pairs and larger image groups. The coarse-to-fine design exposes the model to complementary comparison regimes: coarse-grained groups emphasize robust ranking, best/worst selection, and holistic explanation, whereas fine-grained groups emphasize local artifacts, attribute-level trade-offs, and borderline preferences.

TABLE 1

Statistics of the Stage I comparative-instruction subset in Co-Instruct-359K. Stage I is constructed from two sources, Merge2Compare and Teach2Compare, each covering coarse- and fine-grained comparison with both reasoning and question instruction types.

Category	Merge2Compare				Teach2Compare				Overall
	Coarse-grained		Fine-grained		Coarse-grained		Fine-grained		
	Reasoning	Question	Reasoning	Question	Reasoning	Question	Reasoning	Question	
# Image pairs	9,722	19,448	9,722	19,450	9,998	19,997	10,000	20,000	118,337
# Sets of 3 images	4,795	9,578	4,796	9,575	4,998	9,997	5,000	10,000	58,739
# Sets of 4 images	4,751	9,499	4,757	9,471	5,000	9,999	5,000	10,000	58,477
# Sets of 5 images	4,651	9,358	4,646	9,324	4,998	9,996	5,000	10,000	57,973
# Sets of 6 images	4,619	9,235	4,590	9,224	4,997	9,991	5,000	10,000	57,656
# Data items	28,538	57,118	28,511	57,044	29,991	59,980	30,000	60,000	351,182

3.1.1 Merge2Compare

As shown in Fig. 2, Merge2Compare transforms human-authored single-image quality descriptions into multi-image comparative instructions. We use the 19K images from Q-Pathway [3] as the source pool because each image is associated with a perceptually grounded quality description. The construction consists of two steps.

Candidate group construction (Fig. 2(a)). We first apply Q-Align [4] to estimate the perceptual quality score of each image in Q-Pathway, and then sample image pairs and groups of three to six images according to their score differences. A candidate pair or group is assigned to the fine-grained subset when all pairwise quality-score differences are below 0.5, thereby emphasizing perceptually close samples that require subtle attribute-level discrimination. Otherwise, it is assigned to the coarse-grained subset, where larger quality gaps provide stable ordering cues for ranking, best/worst selection, and holistic comparison. This score-guided construction extends pairwise comparison to listwise supervision and reflects practical scenarios such as candidate selection, quality screening, and near-duplicate comparison.

Instruction merging (Fig. 2(b)). For each candidate group, we prompt the proprietary Qwen3-VL-Plus model [35] to merge the individual Q-Pathway descriptions into a unified comparative instruction. To improve the reliability and ordering consistency of the generated supervision, we provide the Q-Align quality score of each image as an auxiliary score prior in the prompt. This enables the model to ground its comparison in both textual quality evidence and estimated perceptual scores. The prompt requires the generated response to rank the images, identify the dominant quality differences, and provide concise natural-language reasoning. We further convert the merged comparisons into question-based instances so that the resulting supervision covers both free-form explanation and structured comparison. Thus, Merge2Compare preserves the perceptual grounding of human-authored descriptions while extending isolated image evaluation to open-ended multi-image quality understanding.

3.1.2 Teach2Compare

Teach2Compare complements Merge2Compare by generating teacher supervision for images without human-authored comparative descriptions, as shown in Fig. 3. Its purpose is to broaden the content domain, distortion coverage, and comparison patterns beyond those provided by Q-Pathway

descriptions. Specifically, we collect 69K images spanning in-the-wild photographs [6], [36], artificially distorted images [37], and AI-generated content [38], and organize them into image pairs and groups of three to six images for instruction generation.

Coarse-to-fine group construction (Fig. 3(a)). Following the design of Merge2Compare, Teach2Compare adopts a two-level grouping strategy. The coarse-grained subset is formed by grouping diverse images with sufficiently separated predicted quality levels, encouraging robust global judgments across heterogeneous contents and distortion conditions. In addition, the fine-grained subset is constructed from distortion-controlled groups, particularly through KADIS-based synthesis, such that images within a group either share the same distortion type with different severity levels or exhibit different distortion types at comparable severity [39]. This design produces challenging comparisons whose quality differences are subtle, localized, and attribute-dependent.

Teacher-generated comparative reasoning (Fig. 3(b)). We then use Qwen3-VL-Plus [35] as the teacher LMM to generate free-form comparative responses for each group. Each response is required to compare relative perceptual quality, highlight the most relevant quality attributes, and justify the ranking in natural language. Compared with Merge2Compare, this branch introduces richer visual contexts and more diverse image conditions while preserving the comparative reasoning structure required for instruction tuning.

Teacher-generated question answering (Fig. 3(c)). In addition to free-form comparative reasoning, we ask the teacher model to produce structured quality-oriented question-answer pairs for the same image groups. The questions cover multiple formats, including selection, yes/no judgment, attribute-specific comparison, and explanation-oriented queries, with candidate options and reference answers when applicable. This branch strengthens open-ended instruction following by teaching the model to answer diverse comparison prompts rather than only produce rankings. Together, Merge2Compare and Teach2Compare form a Stage I corpus that combines reliable human-derived knowledge with scalable teacher supervision and provides the basis for the score-guided comparative rating instructions in Stage II.

TABLE 2
Template of the score-guided comparative rating instructions used in Stage II.

Prompt. How would you rate the quality of Image 1 and Image 2? Image 1: <image> Image 2: <image>
Response. The quality of Image 1 is <Quality_level_1>, the quality of Image 2 is <Quality_level_2>, Image 1 is <Relation_text> Image 2.
Quality levels: <i>bad, poor, fair, good, excellent.</i> Relation texts: <i>inferior to, worse than, similar to, better than, superior to.</i>

3.2 Stage II: Score-Guided Comparative Rating Instructions

Stage II extends comparative quality understanding to score-guided comparative rating. Its objective is to introduce explicit rating supervision while preserving the comparative reasoning ability acquired in Stage I. Each training instance contains an image pair and a textual response consisting of two absolute-quality labels, one for each image, and one relative-quality label that describes the second image with respect to the first. Unlike language-mediated IQA rating formulations such as DeQA-Score [27], the relative-quality label is not prescribed independently of the underlying score statistics. Instead, it is derived from the pairwise score difference and its uncertainty, yielding a five-level relative-quality vocabulary that is consistent with subjective quality measurements. Whereas Stage I emphasizes open-ended reasoning over multi-image groups, Stage II focuses on pairwise rating instructions that explicitly connect relative judgment with absolute quality estimation.

We construct this subset from the training split of KonIQ-10K [36] by sampling 8K7K image pairs that span diverse contents and quality levels. Each pair is converted into a score-guided comparative rating instruction that jointly supervises absolute quality description and relative quality judgment. The prompt asks the model to assess the perceptual quality of both images and determine their relative quality, while the response assigns a discrete quality label to each image and a relative-quality label to the second image with respect to the first. This design yields instruction-following supervision that is naturally compatible with pairwise comparison and downstream quality prediction.

To instantiate the placeholders in Tab. 2, we convert continuous subjective scores into discrete absolute-quality and relative-quality labels during training.

3.2.1 Training-Time Score-to-Quality Label Conversion

We first map each continuous quality score s to an ordinal quality label. Let m and M denote the minimum and maximum subjective scores in the training dataset, respectively, and define the normalized score as $\bar{s} = (s - m) / (M - m)$. Following the standard five-level quality vocabulary in ITU recommendations [40], we uniformly quantize the normalized score range by

$$L_q(s) = \ell_{\min\{5, \lfloor 5\bar{s} \rfloor + 1\}}, \quad (1)$$

where the ordered absolute-quality vocabulary is

$$\mathcal{Q} = \{\ell_i\}_{i=1}^5 = \{\text{bad, poor, fair, good, excellent}\}. \quad (2)$$

This discretization preserves the ordinal structure of perceptual quality while yielding linguistically interpretable supervision for instruction tuning. It is used to instantiate the textual responses in Stage II, while the rater optimization in Section 4 further exploits the underlying continuous scores through soft labels.

3.2.2 Training-Time Score-to-Relative Label Conversion

We further translate pairwise score differences into relative-quality labels so that the response describes the second image relative to the first in a statistically meaningful manner. For an image pair (x_1, x_2) with mean opinion scores μ_1 and μ_2 , and corresponding score standard deviations σ_1 and σ_2 , we define the signed quality difference and the associated uncertainty as

$$\Delta_{2 \rightarrow 1} = \mu_2 - \mu_1, \quad \sigma_{2 \rightarrow 1} = \sqrt{\sigma_1^2 + \sigma_2^2}. \quad (3)$$

Here, $\Delta_{2 \rightarrow 1} > 0$ indicates that the second image has a higher mean quality score than the first. Under the standard independence assumption, $\sigma_{2 \rightarrow 1}$ characterizes the uncertainty of the pairwise quality difference. We define the ordered relative-quality vocabulary as

$$\mathcal{R} = \{\rho_i\}_{i=1}^5 = \{\text{inferior, worse, similar, better, superior}\}. \quad (4)$$

Following the empirical rule [41], we assign the relative-quality label of the second image with respect to the first by

$$L_r(\Delta_{2 \rightarrow 1}, \sigma_{2 \rightarrow 1}) = \begin{cases} \rho_1, & \Delta_{2 \rightarrow 1} \leq -2\sigma_{2 \rightarrow 1}, \\ \rho_2, & -2\sigma_{2 \rightarrow 1} < \Delta_{2 \rightarrow 1} < -\sigma_{2 \rightarrow 1}, \\ \rho_3, & -\sigma_{2 \rightarrow 1} \leq \Delta_{2 \rightarrow 1} \leq \sigma_{2 \rightarrow 1}, \\ \rho_4, & \sigma_{2 \rightarrow 1} < \Delta_{2 \rightarrow 1} \leq 2\sigma_{2 \rightarrow 1}, \\ \rho_5, & \Delta_{2 \rightarrow 1} > 2\sigma_{2 \rightarrow 1}. \end{cases} \quad (5)$$

This uncertainty-aware formulation maps score differences within one standard deviation to *similar*, moderate differences between one and two standard deviations to *worse/better*, and larger differences to *inferior/superior*. Consequently, Stage II provides a principled bridge from open-ended comparative reasoning to quality-aware score prediction.

4 THE CO-INSTRUCT++ MODEL

Our framework contains two sequentially trained models built on a shared multimodal backbone: Co-Instruct-Assistant, which learns open-ended multi-image quality reasoning from the Stage I instructions, and Co-Instruct-Rater, which is initialized from the assistant and further optimized on the Stage II score-guided comparative rating instructions. The assistant is responsible for comparative understanding and natural-language reasoning, whereas the rater transfers this capability to quality-aware pairwise rating and scalar score estimation. Unless otherwise specified, an input instance is denoted by an image set $\mathbf{X} = \{x_k\}_{k=1}^K$, a textual query u , and an autoregressive target response $y = (y_1, \dots, y_T)$.

4.1 Architecture Overview

We adopt Qwen3-VL-8B [35] as the foundation model. Following the standard multimodal large language model paradigm [42], [43], each input image is first encoded into visual tokens by the vision encoder, after which a vision-language connector maps the visual representation into the language space. We use a vision abstractor as the connector, since it compresses each image into a compact token sequence and thereby reduces the computational burden of multi-image reasoning. This design is particularly important in our setting, where a query may involve up to six images.

To preserve the identity of each image in multi-image prompts, we employ an image-text interleaved input format. Specifically, every image is preceded by an explicit ordinal phrase such as “the first image” or “the second image” before the user query is appended. Compared with directly stacking multiple images ahead of the text prompt, this format makes the correspondence between visual inputs and linguistic references explicit, which is crucial for stable ranking, selection, and explanation in comparative reasoning.

4.2 Co-Instruct-Assistant

Let θ_A denote the parameters of Co-Instruct-Assistant. The assistant is trained on the Stage I instructions to acquire open-ended visual quality reasoning ability. Given a multi-image input $(\mathbf{X}, u)$, it generates the response autoregressively according to

$$p_{\theta_A}(y \mid \mathbf{X}, u) = \prod_{t=1}^T p_{\theta_A}(y_t \mid \mathbf{X}, u, y_{<t}), \quad (6)$$

and is optimized by the standard next-token prediction objective

$$\mathcal{L}_{\text{assist}} = - \sum_{t=1}^T \log p_{\theta_A}(y_t \mid \mathbf{X}, u, y_{<t}). \quad (7)$$

Because Stage I covers both free-form comparative reasoning and question-driven comparison over pairs and larger image sets, this stage equips the model with unified capabilities for ranking, best/worst selection, attribute-level analysis, and natural-language explanation.

4.3 Co-Instruct-Rater

Let θ_R denote the parameters of Co-Instruct-Rater. The rater is initialized from the trained assistant and further optimized on the Stage II score-guided comparative rating instructions. This sequential training strategy preserves the comparative reasoning ability learned in Stage I while adapting the model to produce rating-aware responses. For each image pair (x_1, x_2) , the target response contains two absolute-quality tokens and one relative-quality token, corresponding to the quality of the first image, the quality of the second image, and the relative quality of the second image with respect to the first.

Let $\mathcal{Q} = \{\ell_i\}_{i=1}^5$ and $\mathcal{R} = \{\rho_i\}_{i=1}^5$ denote the ordered absolute-quality and relative-quality vocabularies introduced in Section 3. We use autoregressive cross-entropy, denoted by $\mathcal{L}_{\text{ce}}$, to supervise all response tokens other than the two absolute-quality tokens; this includes the relative-quality

token derived from $L_r(\Delta_{2 \rightarrow 1}, \sigma_{2 \rightarrow 1})$ in Eq. (5). The two absolute-quality token positions are optimized with an ordinal soft-label objective so that adjacent levels are treated as perceptually closer than distant ones.

For a scalar quality score s with subjective standard deviation σ , let m and M be the minimum and maximum scores of the corresponding training dataset. We first rescale s to the canonical five-level range by

$$\tilde{s} = 1 + 4 \cdot \frac{s - m}{M - m}, \quad \tilde{s} \in [1, 5]. \quad (8)$$

We additionally rescale the subjective standard deviation to the same canonical range as

$$\tilde{\sigma} = 4 \cdot \frac{\sigma}{M - m}.$$

Using anchor points $c_i = i$ for $i \in \{1, \dots, 5\}$, we define the soft target distribution over the five quality levels as Using anchor points $c_i = i$ for $i \in \{1, \dots, 5\}$, we primarily construct the soft target from a Gaussian distribution centered at $\tilde{s}$. Depending on the dataset configuration, the unnormalized response at each anchor is computed using either the Gaussian probability density function (PDF) or the probability mass within the corresponding unit-width interval (CDF):

$$g_i = \begin{cases} \phi(c_i; \tilde{s}, \tilde{\sigma}), & \text{PDF,} \\ \Phi\left(\frac{c_i + \frac{1}{2} - \tilde{s}}{\tilde{\sigma}}\right) - \Phi\left(\frac{c_i - \frac{1}{2} - \tilde{s}}{\tilde{\sigma}}\right), & \text{CDF,} \end{cases}$$

where $\phi(\cdot; \tilde{s}, \tilde{\sigma})$ denotes the Gaussian density with mean $\tilde{s}$ and standard deviation $\tilde{\sigma}$, and $\Phi(\cdot)$ is the standard Gaussian CDF. Since the five Gaussian responses do not necessarily sum to one or reproduce the target score, we apply an affine transformation. Let $A = \sum_{i=1}^5 g_i$ and $B = \sum_{i=1}^5 c_i g_i$. The transformation parameters and the nonnegative adjusted weights are

$$\alpha = \frac{\tilde{s} - 3}{B - 3A + \epsilon}, \quad \beta = \frac{1 - \alpha A}{5},$$

$$q_i^G = \max\{\alpha g_i + \beta, 0\}.$$

where ϵ is a small constant for numerical stability. Before nonnegativity clipping, this linear correction enforces $\sum_i (\alpha g_i + \beta) = 1$ and $\sum_i c_i (\alpha g_i + \beta) = \tilde{s}$. After clipping, we measure the residual score-reconstruction error as

$$d = \left| \sum_{i=1}^5 c_i q_i^G - \tilde{s} \right|.$$

Under normal conditions, we use $q_i = q_i^G$ as the ordinal soft target. A fallback is activated when either the normalized standard deviation is too small, $\tilde{\sigma} < \tau_\sigma$, causing an excessively sharp or numerically unstable Gaussian target, or the adjusted distribution cannot accurately reconstruct the normalized score, $d > \tau_d$. Here, τ_σ and τ_d denote the standard-deviation and reconstruction-error thresholds, respectively. In either case, we replace the Gaussian target with the two-nearest-level linear interpolation

$$q_i = \frac{\max\{0, 1 - |\tilde{s} - c_i|\}}{\sum_{j=1}^5 \max\{0, 1 - |\tilde{s} - c_j|\}}, \quad i \in \{1, \dots, 5\}. \quad (9)$$

~~This construction~~ **This fallback** allocates probability mass only to the two nearest levels and degenerates to a one-hot target when $\tilde{s}$ coincides with one of the anchors. For the corresponding model logits z_i , we compute a closed-set softmax over the five absolute-quality tokens:

$$\hat{p}_i = \frac{\exp(z_i)}{\sum_{j=1}^5 \exp(z_j)}, \quad i \in \{1, \dots, 5\}. \quad (10)$$

Let $\mathbf{q}^{(r)}$ and $\hat{\mathbf{p}}^{(r)}$ denote the target and predicted distributions for the r -th image, where $r \in \{1, 2\}$. ~~The ordinal rating loss is defined as~~ **The Stage II rater loss is defined as**

$$\mathcal{L}_{\text{ord}} = \sum_{r=1}^2 \sum_{i=1}^5 q_i^{(r)} \log \frac{q_i^{(r)}}{\hat{p}_i^{(r)}} \quad (11)$$

$$\mathcal{L}_{\text{rater}} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{kl}}.$$

~~Combining the cross-entropy term for the remaining response tokens with the ordinal soft-label term for the two absolute-quality tokens yields the Stage II training objective~~

$$\mathcal{L}_{\text{rater}} = \mathcal{L}_{\text{ce}} + \lambda \mathcal{L}_{\text{ord}}$$

~~where λ is a balancing coefficient. Here, $\mathcal{L}_{\text{kl}}$ denotes the KL-divergence loss between the target and predicted ordinal distributions for the two absolute-quality tokens.~~

4.3.1 Score Estimation during Inference

During inference, Co-Instruct-Rater outputs the probability distribution over the five ordered quality levels. Using the anchor values $\{c_i\}_{i=1}^5$, we estimate the mean and variance of the predicted score distribution as

$$\hat{\mu} = \sum_{i=1}^5 \hat{p}_i c_i, \quad \hat{\sigma}^2 = \sum_{i=1}^5 \hat{p}_i (c_i - \hat{\mu})^2. \quad (12)$$

The expectation $\hat{\mu}$ serves as the final scalar quality estimate, while $\hat{\sigma}^2$ quantifies predictive uncertainty. Because the softmax is restricted to the ordered quality vocabulary, the resulting estimate is less affected by irrelevant textual tokens and remains consistent with the ordinal structure of human quality judgment.

4.3.2 Learning-to-rank Training Strategy

To improve generalization across heterogeneous IQA datasets, we further incorporate a dataset-wise ranking objective when scalar quality annotations are available. The key difficulty in multi-dataset training is that quality scores from different datasets are not directly comparable, even after linear rescaling, because they are associated with different perceptual criteria and score distributions. We therefore impose pairwise supervision only within each dataset, which encourages the model to preserve reliable ranking relationships without forcing inconsistent global score alignment.

For two images x_A and x_B sampled from the same dataset, let their subjective quality scores follow Gaussian distributions $s_A \sim \mathcal{N}(\mu_A, \sigma_A^2)$ and $s_B \sim \mathcal{N}(\mu_B, \sigma_B^2)$. Under the independence assumption, the probability that x_A has better perceptual quality than x_B is given by

$$p_{A>B} = \Phi \left(\frac{\mu_A - \mu_B}{\sqrt{\sigma_A^2 + \sigma_B^2}} \right), \quad (13)$$

where $\Phi(\cdot)$ denotes the Gaussian cumulative distribution function. Using the predicted mean and variance from Eq. (12), the model-induced comparison probability is

$$\hat{p}_{A>B} = \Phi \left(\frac{\hat{\mu}_A - \hat{\mu}_B}{\sqrt{\hat{\sigma}_A^2 + \hat{\sigma}_B^2}} \right). \quad (14)$$

Following the fidelity-based ranking formulation [44], which is closely related to Thurstone’s comparative judgment model [45], we minimize

$$\mathcal{L}_{\text{fd}} = 1 - \sqrt{p_{A>B} \hat{p}_{A>B}} - \sqrt{(1 - p_{A>B})(1 - \hat{p}_{A>B})}. \quad (15)$$

The final optimization objective for Co-Instruct-Rater is

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{rater}} + \gamma \mathcal{L}_{\text{fd}} \\ \mathcal{L} &= \mathcal{L}_{\text{fd}} + \gamma \mathcal{L}_{\text{rater}}, \end{aligned} \quad (16)$$

~~where γ controls the contribution of the ranking-consistency term. Here, $\gamma = 0.05$ controls the contribution of the rater objective.~~ This design enables the rater to retain instruction-following ability while benefiting from additional scalar IQA supervision under a unified probabilistic formulation.

5 THE MICBENCH_v2

To evaluate open-ended multi-image quality reasoning under broader and more challenging conditions, we construct MICBench_v2, an expanded benchmark for the systematic assessment of listwise visual quality comparison. Compared with the previous version, MICBench_v2 substantially enlarges the evaluation space by including image groups of three to six images and by adopting the same coarse-to-fine grouping principle as the Stage I instruction dataset. In total, the benchmark contains ~~3,000~~ **2,000** multiple-choice questions (MCQs) spanning diverse comparison scenarios. Coarse-grained instances emphasize clear quality gaps and stable ranking cues, whereas fine-grained instances focus on perceptually similar or distortion-sensitive samples that require subtle, attribute-aware discrimination. This design enables MICBench_v2 to evaluate both straightforward comparative judgments and more demanding reasoning about nuanced perceptual differences.

5.1 Sourcing Diverse Image Groups

To ensure broad coverage of content, distortion types, and image realism, we construct the image groups in MICBench_v2 from two complementary sources. The first source comprises images from LLVisionQA [2], which are originally drawn from nine IQA-related datasets [6], [36], [37], [46], [47], [48], [49], [50], [51]. The second source contains additional unlabeled images sampled from diverse databases [37], [51], [52], [53], with no overlap with the images used for model training. Based on this image pool, we organize comparison groups of three to six images according to a coarse-to-fine protocol. Coarse-grained groups have sufficiently separated quality levels to assess stable ranking, best/worst selection, and global comparison. Fine-grained groups are intentionally constructed from perceptually close or distortion-sensitive samples, so successful answering requires attention to subtle artifacts, local degradations, and attribute-specific trade-offs. Through this design, MICBench_v2 provides a richer and less biased testbed for evaluating open-ended quality comparison.

		Coarse-grained Questions	Fine-grained Questions
Which		 Question: Which image displays a clear and focused subject? Candidates: A. The third image; B. The first image; C. The second image; D. None of the images Correct answer: A. The third image	 Question: Which image exhibits the highest perceptual quality in terms of sharpness, natural lighting, and minimal compression artifacts? Candidates: A. the third image; B. the fourth image; C. the second image; D. the first image Correct answer: A. The third image
		 Question: Are the clarity levels of all the images consistent? Candidates: A. no; B. yes Correct answer: B. yes	 Question: Is the fourth image significantly blurrier than the first, second, third, and fifth images? Candidates: A. no; B. yes Correct answer: B. yes
Others		 Question: How is the brightness level of the fourth image compared to the second image? Candidates: A. Darker; B. Same brightness; C. Brighter; D. Brightness cannot be compared Correct answer: C. Brighter	 Question: What visual artifact most significantly degrades the interpretability of the subject in the later images compared to the first? Candidates: A. Progressive loss of spatial detail due to increasing blur; B. Compression artifacts causing blockiness; C. Motion blur localized to the rider only; D. Underexposure obscuring shadows Correct answer: A. Progressive loss of spatial detail due to increasing blur

Fig. 4. **Dataset card of MICBench_v2**, illustrating representative question categories and benchmark organization for multi-image quality comparison over image groups containing three to six images.

5.2 MCQ Design and Subjective Ground Truth

Following common practice in LLM and LMM benchmarking [2], [54], [55], [56], we adopt MCQs as the evaluation format of MICBench_v2. Each image group is paired with a human-authored natural-language question and a set of candidate answers, enabling standardized, reproducible scoring while preserving the semantic diversity of open-ended comparison queries. As illustrated in Fig. 4, the questions cover multiple formats, including *Which* questions, yes/no judgments, and other quality-oriented queries about overall quality, ranking, selection, and attribute-specific comparison. The benchmark is not limited to coarse global judgments; it also includes fine-grained questions that probe localized distortions and subtle perceptual trade-offs over larger image sets.

To establish reliable answer keys, we conduct comprehensive subjective testing for all benchmark items. Rather than relying solely on automatic quality scores or expert heuristics, we determine the ground truth of each MCQ from aggregated human judgments collected under a dedicated annotation protocol, followed by expert consistency checking and ambiguity filtering. This procedure is particularly important for fine-grained cases and comparisons involving five or six images, where the reasoning complexity is substantially higher than in conventional pairwise or three-image settings. As a result, MICBench_v2 provides a human-validated, challenging benchmark for open-ended quality reasoning.

6 EXPERIMENTS

This section evaluates Co-Instruct++ on single-image quality understanding, pairwise quality comparison, open-ended multi-image quality reasoning, and scalar IQA. We further

analyze the effects of the proposed comparative instruction data and score-guided rating stage through ablation studies.

6.1 Implementation Details

We adopt Qwen3-VL-8B [35] as the foundation model and fine-tune it on the proposed instruction data. Unless otherwise specified, all images are processed by the official Qwen3-VL image processor with dynamic-resolution resizing, rather than fixed square resizing. Following the default Qwen3-VL configuration, the minimum and maximum pixel budgets are set to $256 \times 32 \times 32$ and $1280 \times 32 \times 32$, respectively, corresponding to 256 – 1280 visual tokens per image under the model’s $32 \times$ spatial compression. The learning rate is set to 2×10^{-5} , and the model is trained for two epochs with a global batch size of 192. All parameters are updated during fine-tuning. The final checkpoint is used for all reported evaluations, and training is conducted on a server equipped with four NVIDIA H20 GPUs. We adopt Qwen3-VL-8B-Instruct [35] as the foundation model and fully fine-tune all parameters on the proposed instruction data (merged_train_9sets.json). Images are processed by the official Qwen3-VL processor with aspect-preserving dynamic resolution. Following our SFT launch configuration, the visual token budget is 512–1280 tokens per image under $32 \times$ spatial compression (image_min_pixels= $512 \times 32 \times 32$ and image_max_pixels= $1280 \times 32 \times 32$). We use AdamW with learning rates of 1×10^{-5} for the LLM and merger and 2×10^{-6} for the vision encoder, cosine decay, and a 3% warmup. Training runs for one epoch with a global batch size of 128 on four NVIDIA H20 GPUs. The last checkpoint is used for all reported evaluations.

TABLE 3

Results on Q-Bench^{PAIR}-A1. Comparative training improves pairwise quality comparison across different question types and low-level concern categories. Reference performance of random guess and non-expert human is included for context. Among compared models, the best result is in **bold**, and the second-best result is underlined.

Model	Question Types			Low-level Concerns				Overall ↑
	Yes/No	What	How	Distortion	Others	In-context Dist.	In-context Other	
Reference Performance								
Random Guess	50.00	32.03	33.16	38.95	41.95	38.69	43.70	39.82
Non-expert Human	78.11	77.04	82.33	78.17	77.22	80.26	76.39	80.12
Open-source Models								
LaViDa	62.65	56.27	54.67	58.56	56.77	52.05	81.43	58.46
Gemma-3-12B-IT	74.46	61.69	70.24	71.17	57.42	71.23	77.14	69.47
CapRL	78.28	79.76	76.61	77.85	83.06	67.74	75.34	78.28
Qwen2.5-VL-7B-Instruct	85.30	76.61	77.51	83.96	68.39	78.99	<u>84.29</u>	80.48
Co-Instruct	86.50	72.20	79.23	80.00	<u>80.62</u>	<u>81.91</u>	74.22	80.18
LLaVA-OneVision-8B	71.57	63.39	61.59	68.65	65.81	60.27	67.14	66.27
Kimi-VL-A3B-Instruct	80.24	79.66	77.85	83.24	73.55	75.80	72.86	79.38
SAIL-VL2	78.80	73.90	78.89	80.18	67.10	76.71	80.00	80.38
Qwen3-VL-8B-Instruct	85.30	74.58	79.24	<u>84.32</u>	67.10	78.08	85.71	79.87
Proprietary Models								
Qwen3-VL-Plus (Teacher)	84.82	72.88	75.09	81.44	72.90	73.06	<u>84.29</u>	78.48
Claude-Sonnet-4.5	71.33	48.47	71.63	61.98	60.00	72.60	71.43	64.66
GPT-5.4	<u>88.92</u>	83.05	85.81	88.47	78.06	88.13	81.43	86.29
Co-Instruct-Assistant	89.16	<u>80.68</u>	<u>79.58</u>	88.47	73.55	79.45	<u>84.29</u>	<u>83.88</u>

6.2 Results on Quality Understanding and Comparison Tasks

6.2.1 Evaluation Protocols

We compare Co-Instruct++ with a broad set of recent multi-modal foundation models. The open-source baselines include LaViDa [57], Gemma-3-12B-IT [58], CapRL [59], Qwen2.5-VL-7B-Instruct [60], Co-Instruct [9], LLaVA-OneVision-8B [61], Kimi-VL-A3B-Instruct [62], SAIL-VL2 [63], and Qwen3-VL-8B-Instruct [35]. We also include three proprietary baselines: Qwen3-VL-Plus [35], Claude-Sonnet-4.5 [64], and GPT-5.4 [65]. The evaluation covers three complementary capabilities: single-image quality understanding, pairwise quality comparison, and open-ended multi-image quality reasoning.

- 1) **Q-Bench^{SINGLE}-A1** [66] evaluates single-image quality understanding with expert-crafted multiple-choice questions spanning diverse question types and low-level perceptual concern categories. We use this benchmark to examine whether comparative instruction tuning transfers to conventional single-image IQA.
- 2) **Q-Bench^{PAIR}-A1** [66] evaluates pairwise visual quality comparison using 1,999 expert-crafted quality-related multiple-choice questions on image pairs. It measures whether a model can identify the higher-quality image under natural-language comparative queries.
- 3) **2AFC-LMM** [39] evaluates fine-grained pairwise quality comparison under a two-alternative forced-choice protocol. LMMs are prompted to select the image with higher perceptual quality in each pair, and the resulting preferences are converted into single-image quality scores using Thurstone’s Case V model [67].

- 4) **MICBench_v2** evaluates open-ended multi-image quality comparison under richer listwise visual context. Unlike benchmarks limited to single images or image pairs, it requires models to reason over groups of three to six images with diverse question types, comparison granularities, and distortion characteristics.

6.2.2 Results on Existing Evaluation Settings

Q-Bench^{PAIR}-A1. Tab. 3 evaluates pairwise quality comparison under multiple question types and perceptual concern categories. Co-Instruct-Assistant achieves an overall accuracy of 83.88%, outperforming the strongest open-source baseline, Qwen2.5-VL-7B-Instruct, by 3.40 percentage points and exceeding non-expert human performance by 3.76 percentage points. It remains competitive with proprietary systems, trailing GPT-5.4 by 2.41 percentage points overall. Across the subsets, it achieves the best result on *Yes/No* questions and a joint-best result on *Distortion*, while obtaining the second-best results on *What*, *How*, and *In-context Other*. These results suggest that the Stage I comparative instructions improve both direct preference selection and context-sensitive perceptual reasoning over paired images.

Q-Bench^{SINGLE}-A1. Tab. 5 further examines whether comparative training transfers to conventional single-image quality understanding. Co-Instruct-Assistant attains 81.20% overall accuracy, surpassing the strongest open-source baseline, Qwen3-VL-8B-Instruct, by 1.33 percentage points and non-expert human performance by 6.89 percentage points. Although GPT-5.4 achieves the highest overall score, Co-Instruct-Assistant obtains the best results on *In-context Distortion* and *In-context Other*, as well as the second-best results on *Yes/No*, *How*, and *Others*. This behavior indicates that

TABLE 4

Results on MICBench_v2 for multi-image visual quality comparison across different question types and varying numbers of input images. Among compared models, the best result is in **bold**, and the second-best result is underlined.

Model	Question Types			Number of Images				Overall (%)
	Yes/No	Which	Others	Three	Four	Five	Six	
Open-source Models								
Gemma-3-12B-IT	82.93	65.27	75.57	73.30	74.05	62.66	64.84	70.92
CapRL	79.27	62.34	75.06	76.50	77.55	60.00	52.34	68.32
Qwen2.5-VL-7B-Instruct	90.49	65.86	87.28	76.78	75.90	67.00	61.66	75.13
Co-Instruct	77.56	71.72	75.57	70.19	73.98	64.34	59.00	73.67
LLaVA-OneVision-8B	85.37	60.42	81.17	75.70	73.94	63.00	56.34	69.62
Kimi-VL-A3B-Instruct	87.99	73.46	89.20	78.17	77.60	75.00	68.56	79.54
SAIL-VL2	72.20	48.03	73.28	66.53	69.36	50.34	49.66	57.96
Qwen3-VL-8B-Instruct	84.39	74.31	88.55	78.47	76.88	73.66	70.00	79.18
Proprietary Models								
Qwen3-VL-Plus (Teacher)	88.26	77.66	89.77	83.45	81.77	73.34	72.66	82.21
Gemini-2.5-Pro	86.31	76.70	88.72	82.35	83.24	72.84	73.60	81.02
Claude-Sonnet-4.5	72.44	67.20	82.95	72.38	78.17	59.34	58.34	71.37
GPT-5.4	86.59	77.49	87.79	86.57	76.78	75.34	78.34	81.38
Co-Instruct-Assistant (7132)	88.54	78.83	90.08	85.97	81.40	77.50	76.34	83.03

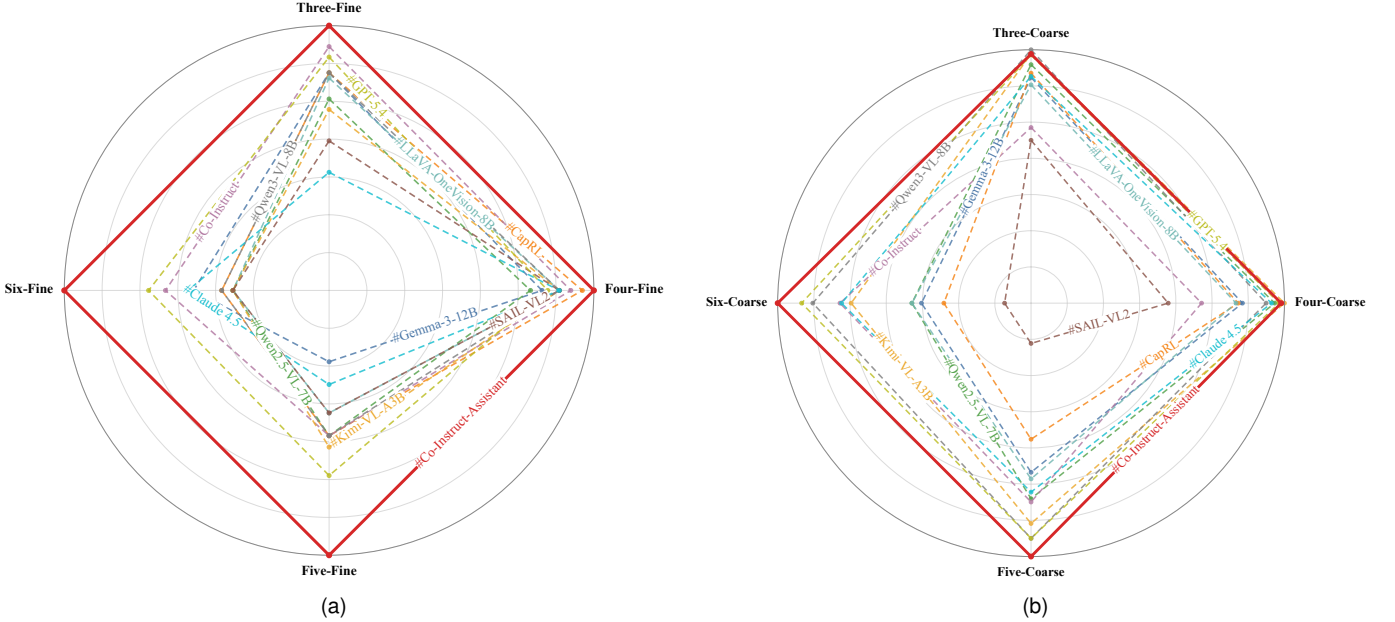


Fig. 5. Fine- and coarse-grained breakdown of multi-image quality reasoning accuracy on MICBench_v2. (a) Fine-grained comparisons. (b) Coarse-grained comparisons. Each radar axis represents a group size of three to six images, and larger radii indicate higher accuracy. Co-Instruct-Assistant exhibits the strongest overall performance profile in both settings.

comparative supervision does not over-specialize the model to multi-image inputs. Instead, learning to identify relative perceptual differences appears to provide a transferable inductive bias for recognizing artifacts, quality attributes, and low-level degradations in single images.

2AFC-LMM. Tab. 6 evaluates whether pairwise quality preferences can be aggregated into reliable global quality estimates. Co-Instruct-Assistant achieves the best average performance, with $\alpha = 0.767$ and $\rho = 0.794$, improving over the strongest competing average result by 0.117 in consistency and 0.047 in correlation. The gains are broad across both synthetic and realistic distortions: Co-Instruct-Assistant obtains the best consistency on five of the six

datasets and the best or joint-best correlation on five datasets. The only clear exception is MM21 in correlation, where Qwen3-VL-Plus remains slightly higher. These results indicate that the learned comparative reasoning is not limited to producing isolated preference answers; it also yields order-consistent pairwise judgments that can be aggregated into robust scalar quality estimates across heterogeneous distortion domains.

MICBench_v2. Tab. 4 reports the multi-image quality reasoning results on MICBench_v2. Co-Instruct-Assistant achieves the highest overall accuracy of 83.03%, outperforming the strongest proprietary baseline, Qwen3-VL-Plus, by 0.82 percentage points and the strongest open-source baseline,

TABLE 5

Results on Q-Bench^{SINGLE}-A1. Comparative training improves single-image quality evaluation across different question types and low-level concern categories. Among compared models, the best result is in **bold**, and the second-best result is underlined.

Model	Question Types			Low-level Concerns				Overall ↑
	Yes/No	What	How	Distortion	Others	In-context Dist.	In-context Other	
Reference Performance								
Random Guess	50.00	28.48	33.30	37.24	38.50	39.13	37.10	37.94
Non-expert Human	82.48	79.39	60.29	75.62	72.08	76.37	73.00	74.31
Open-source Models								
LaViDa	73.54	67.90	60.75	68.16	72.55	70.21	75.67	67.63
Gemma-3-12B-IT	76.46	75.05	69.75	71.02	73.75	71.58	82.13	73.85
CapRL	77.74	80.04	71.40	73.51	77.57	72.26	84.79	76.39
Qwen2.5-VL-7B-Instruct	79.38	82.43	69.34	75.43	77.33	75.00	82.13	77.06
Co-Instruct	81.93	78.74	70.16	74.28	76.37	76.71	84.41	77.12
LLaVA-OneVision-8B	80.29	79.83	72.63	71.98	80.43	75.34	87.07	77.66
Kimi-VL-A3B-Instruct	80.29	82.21	74.49	76.39	78.04	76.37	88.59	79.00
SAIL-VL2	80.66	85.25	73.05	75.82	80.43	79.45	85.93	79.60
Qwen3-VL-8B-Instruct	80.66	83.08	75.93	76.97	78.76	81.16	85.93	79.87
Proprietary Models								
Qwen3-VL-Plus (Teacher)	82.85	75.70	72.22	74.86	75.89	77.74	83.27	77.19
Claude-Sonnet-4.5	81.93	75.70	64.61	71.79	75.66	74.32	77.57	74.38
GPT-5.4	84.85	86.55	74.49	79.65	83.29	82.53	84.03	82.01
Co-Instruct-Assistant	83.94	83.95	75.51	75.82	80.67	82.88	90.87	81.20

TABLE 6

Results on 2AFC-LMM. α denotes the accuracy of consistent binary judgments, and ρ denotes Pearson's linear correlation between MAP-estimated global quality scores and MOSs. The best result is in **bold**, and the second-best result is underlined.

Distortion	Dataset	Qwen3-VL-Plus		GPT-5.4		LLaVA-OV-8B		Qwen3-VL-8B		Co-Instruct-Assistant	
		α	ρ	α	ρ	α	ρ	α	ρ	α	ρ
Synthetic	CSIQ	0.500	0.848	0.617	0.841	0.450	0.753	<u>0.633</u>	<u>0.867</u>	0.778	0.901
	KADID-10K	0.319	0.659	<u>0.635</u>	<u>0.756</u>	0.360	0.590	<u>0.550</u>	0.599	0.744	0.770
	MM21	0.474	0.505	0.552	0.419	0.268	0.272	<u>0.573</u>	0.448	0.652	<u>0.479</u>
Realistic	CLIVE	0.417	0.730	<u>0.658</u>	<u>0.825</u>	0.318	0.594	0.627	0.764	0.802	0.830
	KonIQ-10K	0.442	0.739	<u>0.605</u>	<u>0.749</u>	0.289	0.513	0.590	0.706	0.797	0.878
	SPAQ	0.750	0.907	0.831	<u>0.894</u>	0.757	0.889	0.494	0.805	<u>0.826</u>	0.907
Average		0.484	0.731	<u>0.650</u>	<u>0.747</u>	0.407	0.602	0.578	0.698	0.767	0.794

Kimi-VL-A3B-Instruct, by 3.49 percentage points. It obtains the best results on *Which* questions, other question types, and five-image comparisons, while ranking second on yes/no questions and comparisons involving three or six images. In particular, Co-Instruct-Assistant reaches 77.50% on five-image comparisons, exceeding the strongest competing result by 2.16 percentage points; on six-image comparisons, it achieves 76.34%, behind GPT-5.4 by 2.00 percentage points. These results show that the proposed comparative instruction tuning provides strong and balanced performance across question types and varying comparison-set sizes.

Fig. 5 provides a more detailed view of performance across comparison granularities and group sizes on MICBench_v2. On fine-grained instances, Co-Instruct-Assistant achieves the highest accuracy at every group size, with the clearest separation for five- and six-image groups. These cases combine subtle perceptual differences

with more image identities, requiring the model to retain and jointly compare local artifacts and attribute-level trade-offs. On coarse-grained instances, Co-Instruct-Assistant also exhibits the strongest and most balanced profile. Although several baselines are competitive when quality gaps are clear and only three or four images are compared, their performance declines more visibly as the group size increases. Taken together, the two radar plots show that the gains of Co-Instruct-Assistant are not limited to easier coarse quality distinctions: the proposed comparative instruction tuning improves both stable global ranking and the more demanding discrimination of closely matched images.

6.3 Performance on Image Quality Scoring Tasks

We further evaluate the rating-capable variant, Co-Instruct-Rater, on scalar IQA benchmarks to examine whether open-ended comparative reasoning can be transferred to numerical

TABLE 7

Comparison on image quality scoring benchmarks. Results are reported as PLCC / SRCC on KonIQ-10K [36], KADID-10K [37], PIPAL [68], LIVE-Wild [69], AGIQA-3K [38], and CSIQ [70]. The best result is in **bold**, and the second-best result is underlined.

Method	KonIQ-10K	KADID-10K	PIPAL	LIVE-Wild	AGIQA-3K	CSIQ	Average
NIQE	0.533 / 0.530	0.468 / 0.405	0.195 / 0.161	0.493 / 0.449	0.560 / 0.533	0.718 / 0.628	0.495 / 0.451
BRISQUE	0.225 / 0.226	0.429 / 0.356	0.267 / 0.232	0.361 / 0.313	0.541 / 0.497	0.740 / 0.556	0.427 / 0.363
NIMA	0.896 / 0.859	0.532 / 0.535	0.390 / 0.399	0.814 / 0.771	0.715 / 0.654	0.695 / 0.649	0.674 / 0.645
HyperIQA	0.917 / 0.906	0.506 / 0.468	0.410 / 0.403	0.772 / 0.749	0.702 / 0.640	0.752 / 0.717	0.677 / 0.647
DBCNN	0.884 / 0.875	0.497 / 0.484	0.384 / 0.381	0.773 / 0.755	0.730 / 0.641	0.586 / 0.572	0.642 / 0.618
MUSIQ	0.924 / 0.929	0.575 / 0.556	0.431 / 0.431	0.789 / 0.830	0.722 / 0.630	0.771 / 0.710	0.702 / 0.681
CLIP-IQA+	0.909 / 0.895	0.653 / 0.654	0.427 / 0.419	0.832 / 0.805	0.736 / 0.685	0.772 / 0.719	0.722 / 0.696
ManIQA	0.849 / 0.834	0.499 / 0.465	0.457 / 0.452	0.849 / 0.832	0.723 / 0.636	0.623 / 0.627	0.667 / 0.641
Compare2Score	0.923 / 0.910	0.500 / 0.453	0.354 / 0.342	0.786 / 0.772	0.777 / 0.671	0.735 / 0.705	0.679 / 0.642
Q-Align	0.941 / 0.940	0.674 / 0.684	0.403 / 0.419	0.853 / 0.860	0.772 / 0.735	<u>0.785 / 0.737</u>	0.738 / 0.729
DeQA-Score	<u>0.953 / 0.941</u>	0.694 / 0.687	<u>0.472 / 0.478</u>	<u>0.892 / 0.879</u>	0.809 / 0.729	0.787 / 0.744	<u>0.768 / 0.743</u>
Q-Insight	0.918 / 0.895	<u>0.702 / 0.702</u>	0.458 / 0.435	0.870 / 0.839	<u>0.816 / 0.766</u>	0.685 / 0.640	0.742 / 0.713
VisualQuality-R1	0.924 / 0.915	<u>0.701 / 0.694</u>	0.469 / 0.446	0.873 / 0.841	0.822 / 0.773	0.743 / 0.689	0.755 / 0.726
Co-Instruct-Rater	0.962 / 0.950	0.744 / 0.746	0.557 / 0.512	0.910 / 0.901	0.812 / 0.749	0.758 / 0.707	0.791 / 0.760

quality prediction. We compare with classical no-reference IQA methods, including NIQE [71] and BRISQUE [72]; deep regression-based methods, including NIMA [73], HyperIQA [74], DBCNN [75], MUSIQ [76], CLIP-IQA+ [15], and ManIQA [77]; and recent language- or reasoning-enhanced IQA methods, including Compare2Score [5], Q-Align [4], DeQA-Score [27], Q-Insight [28], and VisualQuality-R1 [29]. The evaluation covers six datasets with diverse content and distortion distributions: KonIQ-10K [36], KADID-10K [37], PIPAL [68], LIVE-Wild [69], AGIQA-3K [38], and CSIQ [70].

6.3.1 Evaluation Protocol

Following standard IQA evaluation practice, we report Pearson’s linear correlation coefficient (PLCC) and Spearman’s rank-order correlation coefficient (SRCC) between predicted scores and human mean opinion scores. PLCC measures prediction fidelity after monotonic score alignment, while SRCC measures whether predicted scores preserve the perceptual ranking of images. The average column reports the arithmetic mean over the six datasets, providing a compact summary of cross-dataset generalization. Higher values indicate better agreement with human subjective judgments.

6.3.2 Main Results

Tab. 7 shows that Co-Instruct-Rater achieves the best average performance, with 0.791 PLCC and 0.760 SRCC, exceeding the strongest competing method, DeQA-Score, by 0.023 PLCC and 0.017 SRCC, respectively. The model achieves the leading results on KonIQ-10K, KADID-10K, PIPAL, and LIVE-Wild, spanning both synthetic and in-the-wild distortions. On KonIQ-10K, Co-Instruct-Rater reaches 0.962/0.950, improving on the second-best result by 0.009/0.009. On KADID-10K, it improves on the strongest baseline by 0.042 PLCC and 0.044 SRCC, indicating greater robustness to controlled synthetic distortions. The improvement is most pronounced on PIPAL, where Co-Instruct-Rater achieves 0.557/0.512 and exceeds the second-best method by 0.085 PLCC and 0.034 SRCC. This result suggests that comparative supervision is particularly beneficial for perceptual

restoration artifacts and fine-grained quality differences. On LIVE-Wild, Co-Instruct-Rater also achieves the best result, improving on DeQA-Score by 0.018 PLCC and 0.022 SRCC.

The results also reveal complementary strengths and limitations. On AGIQA-3K and CSIQ, Co-Instruct-Rater does not achieve the best individual performance, with VisualQuality-R1 leading on AGIQA-3K and DeQA-Score leading on CSIQ. Nevertheless, Co-Instruct-Rater remains competitive on both datasets and achieves the best overall average across all six benchmarks. This pattern suggests that score-guided comparative rating supervision promotes cross-dataset generalization rather than optimizing for a single data distribution, by combining ordinal quality labels, relative-quality supervision, and ranking consistency. Overall, these results support the central hypothesis of this work: comparative reasoning learned from multi-image instruction data can be effectively transferred to scalar IQA prediction.

6.4 Ablation Studies

Tab. 8 evaluates the contributions of the two Stage I comparative instruction sources, Merge2Compare and Teach2Compare, and the KonIQ-10K-based Stage II score-guided comparative rating data. The first four metrics assess quality understanding, spanning single-image, pairwise, and listwise evaluation, whereas the final three measure the transfer of the learned representation to scalar quality prediction. The upper panel isolates the choice of Stage I instruction source while holding Stage II fixed; the lower panel evaluates the effect of Stage II after training with different Stage I sources.

Effect of Stage I comparative instruction. In the upper panel, the Teach2Compare-only variant exceeds the Merge2Compare-only variant on all four quality-understanding metrics: Q-Bench^{SINGLE}-A1 (80.80 vs. 74.92), Q-Bench^{PAIR}-A1 (83.08 vs. 72.87), MICBench_v2 (83.13 vs. 69.62), and 2AFC-LMM (0.759 vs. 0.754). It also performs better on KADID-10K (0.798 vs. 0.698). In contrast, the Merge2Compare-only variant attains higher correlations on LIVE-Wild (0.890 vs. 0.864) and AGIQA-3K (0.740 vs. 0.620). Incorporating both Stage I sources yields the highest

TABLE 8

Ablation study of the training corpus. The validation datasets comprise image quality understanding and quality rating tasks. For each metric, the highest and second-highest results across all reported variants are indicated in **bold** and underlined, respectively.

Training Corpus			Validation Datasets						
Stage II	Stage I		Image Quality Understanding Tasks				Image Quality Rating Tasks		
KonIQ-10K	Merge2Compare	Teach2Compare	Q-Bench ^{SINGLE} -A1	Q-Bench ^{PAIR} -A1	MICBench_v2	2AFC-LMM	LIVE-Wild	KADID-10K	AGIQA-3K
✓	×	✓	80.80	83.08	83.13	0.759	0.864	<u>0.798</u>	0.620
✓	✓	×	74.92	72.87	69.62	0.754	0.890	0.698	0.740
✓	✓	✓	72.91	81.58	82.13	0.727	0.822	0.822	0.674
Stage I		Stage II	Image Quality Understanding Tasks				Image Quality Rating Tasks		
Merge2Compare	Teach2Compare	KonIQ-10K	Q-Bench ^{SINGLE} -A1	Q-Bench ^{PAIR} -A1	MICBench_v2	2AFC-LMM	LIVE-Wild	KADID-10K	AGIQA-3K
✓	×	✓	77.26	74.67	73.22	0.670	0.886	0.698	0.739
×	✓	✓	80.07	80.80	<u>83.18</u>	<u>0.770</u>	0.902	0.747	<u>0.747</u>
✓	✓	×	73.31	83.18	82.48	0.757	0.782	0.791	0.494
✓	✓	✓	81.20	83.88	84.86	0.788	<u>0.901</u>	0.746	0.749

KADID-10K correlation in the upper panel (0.822), but does not improve every metric. These results indicate that Teach2Compare is particularly effective for open-ended quality understanding, whereas Merge2Compare provides complementary supervision that is beneficial for particular rating settings.

Effect of Stage II score-guided rating. The lower panel directly compares models trained with both Stage I sources, with and without the KonIQ-10K-based Stage II data. Adding Stage II improves Q-Bench^{SINGLE}-A1 from 73.31 to 81.20, Q-Bench^{PAIR}-A1 from 83.18 to 83.88, MICBench_v2 from 82.48 to 84.86, and 2AFC-LMM from 0.757 to 0.788. It also increases the correlations on LIVE-Wild and AGIQA-3K from 0.782 to 0.901 and from 0.494 to 0.749, respectively. These gains demonstrate that score-guided comparative rating provides effective calibration from relative quality reasoning to scalar quality prediction. This calibration is not uniformly advantageous across all protocols, as KADID-10K decreases from 0.791 to 0.746, indicating a trade-off on this synthetic-distortion benchmark.

Overall interpretation. No configuration is optimal on every metric. Across the reported variants, the final configuration—which combines both Stage I sources with Stage II rating supervision—achieves the highest scores on Q-Bench^{SINGLE}-A1, Q-Bench^{PAIR}-A1, MICBench_v2, 2AFC-LMM, and AGIQA-3K, as well as the second-highest correlation on LIVE-Wild. The strongest LIVE-Wild result is obtained by the Teach2Compare-only Stage I variant, whereas the strongest KADID-10K result is obtained by the upper-panel configuration using both Stage I sources. Collectively, these findings support the proposed two-stage design: comparative instruction establishes broad quality-understanding capability, and score-guided supervision supplies the calibration required for robust scalar IQA, while preserving distinct contributions from the two Stage I data sources.

7 CONCLUSION

In this work, we formulated IQA as an open-ended multimodal comparative reasoning problem. We constructed Co-Instruct-359K, a two-stage instruction-tuning corpus that combines Stage I comparative-quality-understanding instructions from Merge2Compare and Teach2Compare with Stage II score-guided comparative rating supervision. Based

on this corpus, we developed the Co-Instruct++ series: Co-Instruct-Assistant supports open-ended multi-image quality reasoning, and Co-Instruct-Rater transfers comparative reasoning to scalar quality prediction. We also introduced MICBench_v2, a human-validated benchmark for evaluating listwise quality reasoning over image groups of three to six images. Experiments on single-image understanding, pairwise comparison, multi-image reasoning, and scalar IQA benchmarks demonstrate that comparative supervision provides a unified and generalizable foundation for visual quality assessment.

APPENDIX

PROMPTS FOR STAGE I INSTRUCTION GENERATION

This appendix presents the prompts used to generate the Stage I general-question and multiple-choice-question (MCQ) instructions for Merge2Compare (M2C) and Teach2Compare (T2C). The instruction-generation pipelines support image groups containing two to six images; for clarity and conciseness, we present the two-image case as a representative example, while prompts for larger groups follow the same structure by extending the image descriptions or image inputs and their associated scores. The placeholders <description_1>, <description_2>, <s_1>, and <s_2> are replaced with the descriptions and quality scores of the sampled images. The score information is provided only as an auxiliary quality prior and is explicitly prohibited from appearing in the generated responses.

Merge2Compare Prompts

General-Question Prompt

Merge2Compare: General-Question Prompt

The description of the first image is:
<description_1>
Its score is <s_1>.

The description of the second image is:
<description_2>
Its score is <s_2>.

These two images each have a quality level from 1 to 5. 1 means lowest quality and 5 means highest.

Which image has better quality among the two images, and why? Judge and discuss concisely. Please directly provide your judgement and discussion.

Compare the quality-related attributes and overall quality between the two images briefly. Your reference can be related to specific objects in the two images. Be short and precise.

Do not contain any information that references numeric scores in your response.

MCQ-Generation Prompt

Merge2Compare: MCQ-Generation Prompt

The description of the first image is:
<description_1>
Its score is <s_1>.

The description of the second image is:
<description_2>
Its score is <s_2>.

These two images each have a quality level from 1 to 5. 1 means lowest quality and 5 means highest.

Please generate some yes/no, what, how, and which questions in the multiple-choice question (MCQ) format and corresponding responses on comparing the quality between the first image and the second image in JSON format. Your reply should only contain the dumped list that can be directly loaded via `json.loads`. Do not create any question that references numeric scores.

Please pay more attention to the format of the JSON string:

```
[
  {
    "question": str,
    "options": List[str],
    "correct_option": str (should be one
                        element of the options)
  }
]
```

Generate 3 MCQs if you can.

Teach2Compare Prompts

General-Question Prompt

Teach2Compare: General-Question Prompt

[IMAGE_1 as base64]

[IMAGE_2 as base64]

Each image has an associated quality score (higher means better).

Use these scores for reasoning, but never reveal the numeric values verbatim.

Image 1: score <s_1>
Image 2: score <s_2>

These two images each have a quality level from 1 to 5. 1 means lowest quality and 5 means highest.

Which image has better quality among the two images, and why? Judge and discuss concisely. Please directly provide your judgement and discussion.

Compare the quality-related attributes and overall quality between the two images briefly. Your reference can be related to specific objects in the two images. Be short and precise.

Do not contain any information that references numeric scores in your response.

MCQ-Generation Prompt

Teach2Compare: MCQ-Generation Prompt

[IMAGE_1]

[IMAGE_2]

Each image has an associated quality score (higher means better).

Use these scores for reasoning, but never reveal the numeric values verbatim.

Image 1: score <s_1>
Image 2: score <s_2>

These two images each have a quality level from 1 to 5. 1 means lowest quality and 5 means highest.

Please generate some yes/no, what, how, and which questions in the multiple-choice question (MCQ) format and corresponding responses on comparing the quality between the first image and the second image in JSON format. Your reply should only contain the dumped list that can be directly loaded via `json.loads`. Do not create any question that references numeric scores.

Please pay more attention to the format of the JSON string:

```
[
  {
    "question": str,
    "options": List[str],
    "correct_option": str (should be one
                        element of the options)
  }
]
```

Generate 3 MCQs if you can.

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